

שפות תכנות

תרגיל בית 4

דרה גלזמן 216415331

נבו יצחק 216014639

HW4 - Dry Part

Section A:

1. The symbol ' is necessary to prevent the interpreter from evaluating the arguments *EXP* and *A* before passing them to the *EVAL* function.
Without it, Lisp would evaluate *EXP* and *A* as variables and look for their values but the ' symbol lets them pass as *S*-expressions.
2. The parameter *A* is the a list of key-value pairs so that *EVAL* knows to associate symbols to values in *EXP*.
For example:
(eval 'x '((x . a) (y . b))) means Lisp will search *A* = ((x . a) (y . b)) for the value of x and evaluate *EXP* = x accordingly which means *eval*'s return value will be a, because that is the value for the key x in *A*.

Section B:

The following code should be inserted where we take care of CAR, CDR, QUOTE, etc..

```
1 ((EQ (CAR E) (QUOTE SWAP)))
2 (CONS (CDR (EVAL (CADR E) A)) (CAR (EVAL (CADR E) A))))
```

Section C:

The following code should be inserted where we take care of CAR, CDR, QUOTE, etc..

```
1 ((EQ (CAR E) (QUOTE MAPALL)))
2 ((LABEL MAPALLAUX (LAMBDA (F L)
3     (COND ((NULL L) NIL)
4           (T (CONS (EVAL (CONS F (CONS (CAR L) NIL)) A)
5                     (MAPALLAUX F (CDR L)))))))
6 (CADR E)
7 (EVAL (CADDR E) A)))
```